Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 14 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 2 Class 14 AVSummary

1 UE26ME142A Unit 2 Class 14 AVSummary

Study focus: UE26ME142A_Unit 2_Class 14_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Venkatish from Civil Engineering and the Department of Science and Humanities PS University.
- Two chapters which are there in unit one, chapter one and chapter two.
- In chapter one, we had discussed a few of the definitions and the units of forces.
- In chapter two, we had four topics.
- One as resolution of forces.
- In that you have rectangular components, N and T components, then axes which are not at 90 degrees, then the difference between projection and components.
- That's topic one under chapter two.
- The second topic was on movement.
- We discussed moment, varingance principle, and we have solved problems on moment also.
- That's second topic.
- The third topic was on couple.
- We have seen what is the definition of couple, characteristics of couple, classification of couple, effects of couple, and very importantly, what do you mean by force couple system?
- And then we have solved problems on that.
- The last topic in unit one, chapter two, is resulted.
- We have discussed what is resulted, how do we determine the characteristics of resultant, four characteristics that is magnitude, direction, sense, and point of application.
- To determine the point of application, we have used principle of moments, okay, and then we have solved problems on that.

- To certain extent unit 3 also.
- You have the background of unit one.
- We will now get into unit two.
- And the prescribed textbook that we are using is engineering mechanics statics.
- Very importantly, it is the eighth edition, which is in SI version.
- And the author is Miriam Craig and adopted by Bolton.
- There's one more in this other than equilibrium, we also have trusses.
- We will discuss that later when we take up that topic.
- So first we will discuss about chapter three, which is equilibrium.
- What is equilibrium?
- If the resultant force and the resultant movement acting on a body on which there are several.
- Forces and moments acting, that is a system of forces are acting.
- If the resultant force and moment both of them are zero, then we say that the body is in equilibrium.
- Application of the force would be a rotatory motion which will make this body to rotate.
- It won't move.
- But you can also have a combination of transmitter motion as well as rotatory motion.
- It ensures that there is no transmitter emotion.
- Why I'm saying this?
- Because as you could see, the net force because of a couple is zero.
- But what is the effect of a couple?
- It makes the body to rotate.
- It should not get displaced, it should not rotate also.
- But if we take this, m is equal to 0, or algebraic sum of all the moments of all the forces at any point, at any point should be equal to 0.
- This will ensure that there is no rotatory motion.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

Ensure or to show that the body is in equilibrium or to determine the forces which makes the body to be in equilibrium.

Key details

- What is that?
- The first thing is a free body diagram.
- We have already discussed about free body diagram.
- But since it comes exclusively in this chapter, I am again arbitrating or bringing the concept of a free body diagram.
- As the name suggests, free body diagram is a diagram in which we will have a diagram.
- Such that there is no body in that.

- The diagram will have the forces, the characteristics of forces which are acting on the body, and we were going to remove that body because we are not interested in the body.
- What we are interested is what are the forces that is acting on the body, what is the effect of that.
- That is what is called as a free body diagram, as I suggest, as the name itself suggests, we are making the diagram free of the body.
- Position means the point of application.
- In the solution of problems in mechanics, if how to draw the free body diagram, fifty percent of the problem is solved when it comes to equilibrium chapter.
- But that free body diagram should be clear and sufficient for you to determine the unknowns.
- A few of these supports we have already discussed in our unit one.
- The first type of support is a flexible cable or belt or chain or rope.
- It is a body which can support tensile forces.
- When I say tensile forces, if you have a rope and if you try to pull it, okay, once it is tight, afterwards it will not move.
- For this type of support where you have a flexible cable or a bolt, or sorry, belt, or a chain, or a rope.
- It can resist tensile forces, and that is what is indicated here.
- So this, if I try to pull this it will not move because of this oh.
- And there's a tension here, the internal forces will be acting towards each other and that is what is indicated here.
- If I cut it here and look at this, I will get the force to be this.
- There will be only one force with no line of action.
- If you are looking at what is the force acting on the in the row, then it is internal.
- If you are looking at what is the force causing the tension is the external force.
- The second one is a smooth surface.
- A smooth surface is one in which you will have one reaction with known line of action.
- Which will always act perpendicular to the surface.
- This is the surface.
- And this is the reaction that we get and this angle is always ninety degrees and the sense is going to be opposite to the body weight that is acting.
- The body weight will be acting here, so you will have the normal reaction acting here.
- There won't be any friction when it comes to smooth surface.
- Or the frictional force that develops because of the smooth surface would be quite less when compared to the force that is acting.
- The next category is a rough surface.
- You have already studied about this.
- If the roughness of the body which are in contact with each other is quite large.
- Then it will prevent the body from moving tangential to the surface, and that we call it as frictional force.
- In addition to this frictional force, we will also have a normal reaction.

Example or application from the lecture

- Let us look at that.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

to the normal.

Key details

- We can represent these two by one resultant force, but the direction of this resultant force depends upon the magnitude of F1N.
- The next category of the type of support is a roller support.
- Four different ways of representing the rollers.
- You can also have a roller vertical or inclined also.
- with known line of action, which will always be acting perpendicular to the line of rollers.
- In this case the line of roller is horizontal in all the four cases.
- You can also have vertical if the rollers are to the left, then you will have the support reaction acting to the right.
- If you have the rollers.
- To the right, then you will have the reaction to the left.
- The next one is a freely sliding guide.
- In this case, you will have one reaction only.
- In this case, you can see that the sliding guide can move horizontally, it cannot move vertically.
- The reaction can be upwards or downwards.
- Along the horizontal, it can move only this much.
- After which, if it moves till here, then it cannot move.
- Then you will have a reaction in the horizontal direction also.
- That is why it is called as freely sliding train.
- Only till that position.
- The horizontal support will not be there.
- After that, you will have the horizontal support.
- But however, here what is indicated for this position is there will be only one normal reaction, normal reacting vertically.
- Continuing the different types of supports, the next one comes.
- Pin connection, look at how the pin connection is indicated.
- There is a clear difference between a pin connection and a roller connection.
- In case of rollers, you will have roller here and then you will have a the surface.
- Whereas pin connection, directly the triangle is resting on the surface.
- The door frame or the door itself is fixed to the frame of the door.
- In case of hinges, you will have two reaction.
- No, you cannot move the door upwards or downwards.
- Toward the frame, or you cannot pull it out of the frame, but you can rotate the door.
- That is permitted in case of the door.

- That is what the hint support does.
- It will provide you two reactions.
- It will permit you to rotate.
- In the plane opposite to this two reaction that is coming.
- If this is X and Y, it will rotate along the Z axis.
- No, you have a door with a stopper.
- Doe stopper, we are we know the door stopper.
- Because the doorstopper, it will prevent the door from closing.
- Then we will have the moment also.
- That is why you can see here.
- Pin not free to turn.
- Where you will have restriction for rotation.
- The next one is a fixed or a built-end support.
- This is welded connection, this is fixed.
- The sense of this, even here the sense of this reactions can be upwards or downwards, right side or left side, clockwise or anti-clockwise based on the force that is acting on the body.
- Then we have also looked at gravitational force earlier.
- Always the gravitational force will be acting downwards.

Example or application from the lecture

- The best example that I can give you is the door.
- Hinges are the best example for pin connection.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

Okay, you cannot round it off to ten because if you round it off to ten then you will not get the correct answer what is mentioned in the textbook.

Key details

- Next other one important type of support is a spring connection.
- The spring, in case of spring connection, the spring will allow it to displace or it will allow it to elongate or compress, but with certain force.
- Which defines what should be the force that you need to apply for unit elongation or compression.
- That is what is called as stiffness K.
- elongation or compression that I have to give, then that we can find it as F is equal to K times the elongation or compression.
- Along with this, we also have one more, what is called as torsional spring action, where instead of a linear spring, this is a linear spring where you can elongate or compress.
- In case of torsional spring,.
- You can you have to apply a movement so that it rotates.
- That is m by theta.

- Yes, KS or KT torsional spring is equal to what is the moment required for an unit displacement.
- You can see here the body as such is removed, only the outer edges are intact.
- Then the supports replaced by their respective reactions.
- Here it is an inch support, so you have two reactions, here it is a roller support, so you have one reaction.
- If you don't understand, you can always come and talk to me later.
- First one now here you can see there is a force acting here.
- This is a cable, so there will be one reaction and they have indicated that tension.
- Here they have kept a mass here.
- So because of this mass there will be a force which is m times g.
- Here you have a.
- Hinge support which is not indicated here.
- You have to identify what is incomplete here.
- And according to that you should complete the free body diagram.
- I hope you can do that.
- Next, this table gives you a free body diagram, which is wrong or incomplete.
- You can see here it is wrong or incomplete.
- I'll explain the first one.
- You have a roller inclined on an inclined surface, or kept on an inclined surface, there's a force of P that is applied.
- What is wrong?
- The force P is acting here, that is exactly right, P is correct.
- If this is a smooth surface, so you have one reaction normal to the surface, that is also correct.
- Mg is the force because of the sulfate of the roller.
- That is because of gravitation.
- It should act vertically down.
- You have to draw the free body diagram for all this, sorry, eight diagrams, all this eight diagrams.
- I will explain the first one.

Example or application from the lecture

- But you can see here they have given Mg.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

Suspended by vertical cable at A and supported by rough inclined surface at B.

Key details

- This is a cable.

- Then this I will remove it and replace it by a force acting upwards because it's tension, the internal forces will be acting upwards.
- Whereas here there will be a normal reaction acting upwards, then there will be a frictional force acting along this surface and it is upwards.
- If you have any doubts you can meet.
- We will stop it here.
- We'll continue in the next class.
- My name is Dr.
- Sv Vength from Civil Engineering under Department of Science and Humanities, PS University.
- This is class number eighteen.
- Next we will have class number nineteen.

Example or application from the lecture

- Let's meet in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 2_Class 14_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 14_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 14_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 14_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.
4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.

9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00002	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00003	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00005	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

C Revision and retrieval practice